Merge 2048 - Game UI/UX Audit Report

Audit date: 2026-10-16

Scope: Read-only source inspection, existing automated tests, existing production-build result, and available local server response.

Verdict: NOT VERIFIED - there is not enough live UI evidence to make a reliable release-readiness claim.

1. Executive summary

Merge 2048 is a browser-based tile-merging puzzle built with React 18, TypeScript, Vite, and CSS. It provides seven game modes, configurable 4x4/5x5/6x6 boards, local progress storage, and touch, pointer, and keyboard move handling. The codebase has a coherent theme system and a good set of explicit game-state screens. The existing unit suite passed all 20 tests across three test files, and the production build completed successfully during the preceding launch troubleshooting. The local preview endpoint returned HTTP 200 with the page title "Merge 2048."

The running interface could not be visually inspected or interacted with in this audit session: the available browser automation blocked access to the localhost page. No screenshots, viewport measurements, browser-console observations, or interaction-level results are claimed. The score below is a **provisional source-based assessment**, not a visual usability certification.

Five confirmed code-level issues merit attention: restricted modes still expose undo/power-ups, undo restores only part of the game state, gameplay registers unload listeners that its cleanup cannot remove, daily streaks increment without checking the completion date, and the DOWN movement path reads frozen tiles from the wrong column. A further issue affects Time Attack completion statistics. These findings warrant regression tests before release.

Provisional UI/UX score: 6.2/10.

Functional verification: Partial. 20/20 automated tests passed; launch/build checks passed; manual UI acceptance is blocked.

Readiness: NOT VERIFIED.

2. Game overview

Identity and technology

Item	Finding
Name	Merge 2048 (from document title, splash, and main-menu heading)
Genre	Number-tile merge / 2048-style puzzle
Stack	React 18, TypeScript, Vite 6, CSS, lucide-react icons, Vitest
Runtime	Web browser; no native mobile or desktop packaging found
Persistence	Browser `localStorage` with in-memory fallback; no server API found
Network dependencies	Google Fonts stylesheet in `frontend/index.html`; core game logic is local
Board sizes	4x4, 5x5, 6x6
Modes	Classic, Endless, Time Attack, Moves Challenge, Bomb, Ice, Daily Challenge

Repository structure: application shell and navigation are in `frontend/src/App.tsx`; screens in `frontend/src/components/screens/`; board and controls in `components/board/` and `components/common/`; rules in `frontend/src/game/`; save logic in `frontend/src/utils/gameStorage.ts`; shared types in `frontend/src/types/game.ts`; styling and themes in `frontend/src/styles/` and component CSS. The root `package.json` forwards dev/build commands to `frontend/`.

Screens and states identified in source

Splash, first-run tutorial, main menu, Moves Challenge selection, Daily Challenge overview, gameplay, best scores, statistics, How to Play, settings, About, first-use mode-introduction dialog, pause dialog, restart confirmation, game-over dialog, challenge-complete dialog, challenge-failed dialog, and reset-progress confirmation.

This list is source-derived. The accessible screens and visual transitions were not manually confirmed in the browser.

Gameplay rules and controls

Standard moves slide tiles in one of four directions. Matching non-bomb tiles merge once per move; the resulting value contributes to score. A valid move spawns a 2 or 4 tile. Classic ends when no moves remain and permits continuation after 2048; Endless removes that target; Time Attack starts at 60 seconds and awards time bonuses for merges. Moves Challenge and Daily Challenge use target/move-limit completion states. Bomb mode uses fuse timers, defusal bonuses, and area clearing; Ice mode uses frozen tiles that require adjacent merges to thaw.

Input is wired for arrow keys and WASD, pointer swipes, touch swipes, and an optional on-screen D-pad. Mouse/touch tile selection is also used by the hammer power-up. Undo, hammer, and shuffle are shown in the action bar. The game stores settings, records, statistics, tutorial/intro flags, and an active game in local storage when available.

Source references

- Navigation, persistence setup, and screen selection: `frontend/src/App.tsx`.
- Mode definitions and challenge targets: `frontend/src/game/modeManager.ts`.
- Tile operations and merge logic: `frontend/src/game/engine.ts`.
- Special tiles and daily puzzle generation: `frontend/src/game/specialTileManager.ts`, `frontend/src/game/dailyChallengeGenerator.ts`.
- Pointer/touch/keyboard bindings: `frontend/src/hooks/useSwipe.ts`.
- State persistence: `frontend/src/utils/gameStorage.ts`.

3. Screen-by-screen UI audit

Screenshots were not captured; see `SCREENSHOTS/README.md`. The following design assessments use component structure and CSS only.

Screen/state	Source-based assessment	Risks / recommendations
Splash	Branded title, tile illustration, short loading animation, 1.8-second transition.	Verify first-launch timing and reduced-motion behavior; no skip control is visible in source.
Tutorial	Dedicated first-run sequence with next/complete action.	Runtime flow, progress affordance, keyboard operation, and text legibility remain unverified.
Main menu	Clear mode cards, board-size selector, score summary, and secondary destinations.	Seven vertically stacked mode cards may require substantial scrolling on short screens. Confirm which actions are above the fold at 360x800.
Challenge selection	Seven levels with unlock/progression state and back navigation.	Inspect disabled/unlocked semantics, progress clarity, and card tap targets at runtime.
Daily Challenge	Date-specific objective, streak/history display, and start action.	Verify empty/history states, date rollover, and streak messaging; the date logic has a confirmed flaw (BUG-04).
Gameplay	Score/HUD, board, action bar, optional D-pad, mode hints, toasts, and game-state overlays are separated into components.	The board has no explicit grid/cell accessibility semantics. Action availability and undo consistency have confirmed defects (BUG-01/02).
Best Scores	Dedicated records screen with back control.	Runtime empty-state hierarchy and data formatting not inspected.
Statistics	Dedicated aggregate-statistics screen.	Validate totals against completed games, especially Time Attack (BUG-05).
How to Play	Dedicated instruction screen accessible from menu/settings.	Compare every instruction against actual controls and special-mode rules.
Settings	Theme and board-size selectors; sound, music, vibration, and animation toggles; links and reset confirmation.	Toggle input names may be empty to assistive technology because visible setting text is not associated with each checkbox (BUG-06). Reset itself was not activated.
About	Dedicated information screen with back navigation.	Content and external links were not runtime-reviewed.
Mode intro	Explanatory modal for Moves, Bomb, Ice, and Daily modes.	Confirm focus handling, Escape behavior, and small-screen fit; no dialog semantics or focus management were found in source.
Pause / restart confirmation	Resume, restart, main-menu actions and a separate restart confirmation.	No keyboard/focus behavior could be tested; confirm-modal action does not destroy saved progress until confirmed.

Screen/state	Source-based assessment	Risks / recommendations
Game over / challenge complete / challenge failed	Result metrics with retry/menu/next-level actions.	Verify timer expiry and completion statistics (BUG-05), plus modal focus/announcement behavior.

4. UI/UX scorecard

Scores below are provisional assessments from source structure/styles and automated checks. Actual rendered contrast, clipping, and perceived polish were not observed. The average is over categories 1-11; category 12 is that calculated average and is excluded to avoid circular scoring.

Category	Score / 10	Evidence, weakness, and recommended improvement
Visual design	7.0	A consistent five-theme token system, tile colors, icons, radii, and shadows exists (`styles/variables.css`). Actual visual hierarchy and contrast in all themes are unverified. Inspect each theme in-browser and measure text contrast.
Layout and alignment	6.0	Screens/components use flex/grid and a centered 520px app frame (`styles/index.css`). No rendered alignment checks were possible; gameplay uses `space-evenly` with hidden overflow, which should be checked for short landscape viewports.
Typography and readability	6.0	Outfit/Plus Jakarta Sans are declared with system fallbacks. Several labels in the action bar and menu use 0.55-0.68rem sizes. Check readability at device scale and preserve zoom where practical.
Navigation and usability	8.0	Explicit screen routing, back actions, main-menu destinations, pause, retry, and confirmation flows are present. Runtime discoverability and keyboard-only traversal remain unverified.
Gameplay clarity	7.0	Tutorial, mode cards, mode intros, and bomb/ice HUD hints provide layered instruction. Validate copy consistency and help for power-ups; mode restrictions are not respected by the action-bar state (BUG-01).
Interaction feedback	7.0	Merge/milestone toasts, sound, haptics, floating scores, confetti, and explosion effects are wired. No live feedback or browser audio-permission behavior was inspected.
Mobile responsiveness	6.0	CSS includes 440px, 380px, and 360px breakpoints, safe-area insets, and touch gestures. Required 360x800/390x844 layouts were not captured.
Tablet and desktop responsiveness	5.0	The UI remains a centered app-like column capped at 520px. No tablet/desktop-specific layout is evident. Test 768x1024, 1366x768, and 1920x1080 before making a responsive claim.
Accessibility	4.0	Semantic buttons and several `aria-labels` exist. Board cells lack grid semantics; settings checkboxes are not associated with their visible text; dialog focus, focus visibility, reduced motion, and live announcements are not evident.
Animation and polish	7.0	CSS and React effects include tile pop/merge, toast, explosion, and confetti feedback, plus an animation toggle. Reduced-motion preference support was not found.
Performance and responsiveness	5.0	A small client-side board and pure rule functions should be manageable. Gameplay registers unload listeners per game-state change and removes different anonymous callbacks, creating a confirmed listener leak (BUG-03). No timing or memory measurements were collected.
Overall UI/UX quality	**6.2**	Mean of categories 1-11: 68 / 11 = 6.18, rounded to 6.2. This is provisional and does not substitute for viewport/browser validation.

5. Functional verification matrix

The automated test command was `npm test --prefix frontend`: **3 files passed, 20 tests passed, 0 failed**. It covers engine mechanics, special tiles/daily generation, and storage utilities; it is not an end-to-end browser suite. Production build passed during the previous launch troubleshooting (`npm run build --prefix frontend`; Vite transformed 1,632 modules). The preview endpoint returned HTTP 200 and an HTML title. Those checks do not prove the game rendered or that user flows work.

Feature	Test performed	Result	Evidence	Notes
Application startup	Request local preview page	Pass (server response)	HTTP 200, title `Merge 2048`	Browser rendering was not observable in this audit.
Main menu	Inspect and use controls	Blocked	Main-menu source only	Browser interaction was denied by the available UI automation policy.
Core gameplay	Start and make representative moves	Blocked	Engine unit tests only	No end-to-end board session or move sequence was executed.
Merge and scoring rules	Existing unit tests	Pass (unit scope)	`engine.test.ts`, 7 tests	Covers merge scoring, invalid move, no triple merge, valid-move detection, and highest tile.
Special modes	Existing unit tests	Pass (partial unit scope)	`phase3.test.ts`, 7 tests	Covers bomb/ice helpers and deterministic daily generation; does not exercise React mode transitions.
Pause/resume and restart	UI interaction	Blocked	Source flow located	No live interaction performed.
Win/loss and challenge progression	UI interaction	Blocked	Source flow located	No complete game was played; challenge and timer outcomes remain unverified.
Save/load persistence	Existing storage tests	Pass (unit scope)	`gameStorage.test.ts`, 6 tests	Browser reload persistence and real browser storage were not exercised.
Responsive layouts	360x800, 390x844, 768x1024, 1366x768, 1920x1080	Blocked	CSS breakpoints reviewed	No genuine screenshots or overflow measurements available.
Production build	Existing build command	Pass	Vite build output: 1,632 modules transformed	Build was run during launch troubleshooting immediately before this audit.
Keyboard, pointer, touch	Manual input checks	Blocked	`useSwipe.ts` source reviewed	Wiring found; behavior and browser/device differences not tested.
Settings and theme persistence	UI toggle and reload	Blocked	Source/storage unit tests	No theme toggles or real reload tested.
Console, network, assets	Browser devtools inspection	Blocked	Source references Google Fonts and local logo	Console and network request status unavailable.

Manual acceptance tally: 2 server/build checks passed; 3 unit-test feature groups passed; 10 manual UI checks blocked. Automated suite tally is reported separately above (20 passed, 0 failed).

6. Bug and issue register

Confirmed defects and source concerns are detailed in `BUGS_AND_ISSUES.md`. Highest impact: rules that disable undo/power-ups are not enforced in the rendered gameplay controls; undo restores only part of the game state; frozen tiles are read from the wrong column during DOWN movement; and daily streak/date handling is not guarded.

7. Responsive UI report

CSS has responsive rules at 440px and 360px in the shared shell, game header, action bar, plus 380px rules for the board and D-pad. The board is sized with CSS grid and a calculated cell width. The top-level app is centered and capped at 520px. These are source observations, not viewport test results.

Viewport	Result	Evidence / limitation
360x800	Blocked	360px breakpoint exists; no actual render.
390x844	Blocked	440px rules apply; no actual render.
768x1024	Blocked	App max-width suggests centered narrow layout; no render.
1366x768	Blocked	No desktop-specific reflow evident; no render.
1920x1080	Blocked	App remains capped at 520px by source; no render.

Recommended visual checks: short-height landscape gameplay, menus with all seven modes, modal height, safe-area behavior, smallest board labels, and action-bar tap targets.

8. Performance and technical report

- Build: passed as noted above; no build error remained in the successful run.
- Tests: 20/20 passed across 3 files, in 3.16 seconds reported by Vitest.
- Browser startup: the local preview endpoint returned HTTP 200; DOM render and browser console were not observed.
- Architecture: client-side React SPA; no game backend/API detected in the inspected source.
- Persistence: versioned JSON envelope stored via localStorage with in-memory fallback. Storage unit tests pass; browser persistence and quota behavior were not tested.
- External resources: `frontend/index.html` references Google Fonts. Offline gameplay can function from local code, but those web fonts are not guaranteed offline.
- Security posture: `frontend/vite.config.ts` binds the dev server to `0.0.0.0` and sets `allowedHosts: true`. This is a development-server exposure concern if used on an untrusted network; no exploit was tested. Bind to loopback and restrict allowed hosts when local-only use is intended.
- Performance measurements, memory profiling, network failures, and leak reproduction were not performed. The listener leak is inferred directly from source registration/removal identity.

9. Accessibility report

Automated accessibility scan: Not run. **Manual screen-reader/keyboard checks:** Blocked.

Source-level positives: native HTML buttons are widely used; many icon-only controls have accessible names; the game supports global arrow/WASD input and optional visible D-pad controls.

Source-level risks: board and tile elements are `div`/`span` structures without grid/cell semantics or announced values; several settings checkboxes are not labelled by their visible setting text; dialogs have no evident `role="dialog"`, `aria-modal`, focus trap, initial focus, or focus restoration; no `prefers-reduced-motion` handling was found; focus-visible styling and live announcements for score/timer/toasts were not found. High contrast should be measured for muted labels and all five themes.

10. Final recommendations

1. **Priority 1 - gameplay integrity:** enforce `ModeConfig.allowsUndo` and `allowsPowerUps`; include all mutable gameplay fields in undo snapshots or define exactly what undo reverses; repair DOWN frozen-tile scanning; reconcile daily streaks against the last completed date; record Time Attack completion statistics on timer expiry.
2. **Priority 2 - accessibility:** label settings inputs, expose the board and tiles semantically, and add modal focus management and screen-reader announcements.
3. **Priority 2 - responsive QA:** capture the five requested viewport sizes, test short landscape height and touch targets, then adjust only after measured issues are confirmed.
4. **Priority 3 - lifecycle/performance:** use stable event-handler references or avoid per-state unload listeners; profile long sessions to confirm cleanup.
5. **Priority 3 - offline/development setup:** provide a local font fallback if offline visual consistency matters; restrict dev-server host exposure for local-only use.
6. Add targeted regression tests for undo completeness, restricted modes, frozen tiles moving DOWN across every column, repeated daily completion/date gaps, Time Attack expiry stats, and listener cleanup.

11. Final verdict

NOT VERIFIED. The code builds, the local page responds, and all 20 existing unit tests pass. However, the live game UI and requested viewport/device flows were not observed, no screenshots were captured, and the source review identified multiple gameplay and accessibility concerns. A release-readiness claim requires interactive browser verification and regression coverage for the confirmed defects.

Evidence and limitations

This report separates code-derived findings from executed test/build/server checks. Existing workspace changes were preserved; no game source, configuration, dependency, asset, browser storage, or saved progress was modified or reset for this audit. Only new audit artifacts were created. Source inspection cannot verify rendered behavior.